A Simulation-First Framework for Predictive Supervisory Control of CMP under Electrical and Ultrapure-Water Disturbances: Public Virtual Metrology, Physics, Coupling, Early Warning, and Attribution

Living manuscript for the Semiconductor CMP Predictive-Control PoC

Journal-neutral draft; author list and affiliations pending

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Evidence boundary. This living paper reports verified data provenance, leakage-safe PHM average-MRR virtual metrology, synthetic plant/CMP/coupling evidence, the frozen WP12 simulation-only early-warning experiment, and WP13 conditional synthetic root-cause classification. The PHM target unit is source-undeclared and the selected tree's primary intervals undercover. WP12 has only three independent held-out events and fails under important shifts. WP13 uses a diagnostic positive-warning token, and delay/dropout collapse its availability. Controller efficacy and safety-filter outcomes remain unavailable. No real-fab, physical-defect, yield, equipment-protection, or production-control conclusion is made.

Abstract

Chemical mechanical planarization (CMP) depends on process settings, consumable condition, and utility support. This work develops a simulation-first proof of concept for studying whether an eventual predictive supervisory layer can warn of and mitigate simulated material-removal-rate (MRR) excursions caused by electrical and ultrapure-water (UPW) disturbances. The intended chain is electrical disturbance, UPS and variable-frequency-drive response, motor and pump response, UPW hydraulic and thermal response, an explicitly declared CMP boundary, and finally a change in simulated MRR. The PHM 2016 CMP archive was locally supplied, integrity-checked, selectively extracted, and audited. A leakage-safe offline comparison uses 311/275 precedence-retained test/final-validation records from mutually disjoint wafers in the source-native target scale. A tree selected before holdout access attains MAE 3.185/3.395 and R^2 0.975/0.958, reducing MAE by 89.2%/88.6% relative to a mean baseline. Its nominal 90% split-conformal intervals cover only 83.9%/84.7%, below a frozen 85% interpretation floor, and a chronological stress is dominated by one preregistered extreme label. The physics-plus-residual hybrid is significantly worse than the tree, so no hybrid-improvement claim is supported. A reduced-order CMP subsystem implements explicit process modes, signed rotary kinematics, generalized pressure–velocity exposure, dynamic consumable and slurry states, an interface energy balance, and cumulative removal. Its nominal reference is 100 nm/min at 30 kPa and passes limiting-case, dimensional, energy-residual, quadrature, replay, and timestep-refinement checks. A typed stateless coupler implements no connection, dressing-water support, a conditional thermal loop, and an explicitly synthetic slurry-support structure. A healthy-UPS 25%, 400 ms voltage sag is an exact coupling negative control. In a separately declared synthetic stress test, a 500 J degraded-UPS interruption during dressing drives pump and UPW support to zero, reduces end-of-dress pad activity by 5.36%, and reduces mean MRR during later polishing by 3.19% relative to an otherwise identical no-connection run. This delayed effect is mediated by pad state, not a direct UPW-to-MRR multiplier. Global and mismatch analyses show material dependence on synthetic topology and coefficients. These results establish a testable synthetic plant and coupling basis. A separately frozen warning target predicts whether active-POLISH MRR will leave a paired-reference ±5% band persistently for 0.25 s within a 3 s horizon. Across 24 held-out synthetic runs containing only three independent events, logistic regression achieves PR-AUC 0.9904 and the gradient-boosted model 0.9130; both detect 3/3 events with 2.71 s median lead. However, both alarm severely when upstream events are replayed under no CMP connection, and high synthetic noise degrades discrimination and uncertainty coverage. The warning result is therefore limited to the named primary ensemble and does not establish predictive-control efficacy, real-tool calibration, or topology-independent applicability. A separately frozen diagnostic study compares always-unknown, residual-rule, multinomial-logistic, and hybrid root-cause estimators on 66 held-out whole runs. The preregistered hybrid attains 0.8636 accuracy and macro recall, 1.0000 unknown recall and selective accuracy, and 0.8500 known-cause coverage; all nine errors are abstentions. Rule only scores higher at 0.9242, but is not substituted after TEST access. Severe delay and dropout collapse diagnostic availability toward unknown. This is simulator-conditional classification, not experimental causal proof or permission to act.

Keywords: chemical mechanical planarization; mate-

rial removal rate; ultrapure water; reduced-order modelling; virtual metrology; early warning; root-cause attribution; supervisory control; sensitivity analysis; simulation

1 Introduction

CMP combines mechanical contact, relative motion, slurry chemistry, pad condition, and heat transfer to remove material. Pressure–velocity proportionality provides a classical starting point [1], but process dynamics and conditioning history make an instantaneous algebraic model insufficient for disturbance and control studies. Kinematics affect local exposure [2]; pad wear and conditioning introduce state memory [3–5]; and slurry and temperature can alter the process through distinct pathways [6–8].

The central research question is whether a predictive supervisory-control layer can detect an upcoming simulated CMP excursion caused by electrical and UPW disturbances and reduce it relative to no intervention and fixed-threshold control without violating predefined simulation constraints. Answering that question requires more than fitting a predictor. The upstream plant must obey energy and hydraulic balances, the connection into CMP must be explicit, the true latent state must remain distinct from delayed and corrupted observations, and controller comparisons must be paired on identical disturbances and seeds.

This manuscript is intentionally staged. The completed contribution comprises a leakage-safe PHM average-MRR comparison, a scientifically bounded synthetic plant, a declared utility-to-CMP connection with null controls and sensitivity, an open-loop early-warning experiment with disjoint run-level fitting/calibration roles, and conditional synthetic root-cause attribution with mandatory abstention. Predictive control, safety filtering, and the integrated comparison remain protocols until their acceptance evidence exists.

The present contributions are:

1. a data and claims contract that separates measured process output, simulated physical state, observed sensor state, process excursion, quality-risk proxy, physical defect, and yield outcome;
2. a target-blind whole-wafer split, six-family source-native virtual- metrology comparison, grouped uncertainty analysis, and explicit label, feature-proxy, continuity, and chronological sensitivities;
3. a reduced-order utility and CMP simulator with conserved UPS energy, hydraulic mass balance, explicit process modes, cumulative removal, and causal observation clocks;
4. a typed topology map that replaces an earlier direct UPW-to-MRR gain with bounded dressing-water, thermal, synthetic slurry-support, and null structures;
5. negative-control, positive-chain, numerical, structural, local, global, and mismatch evidence for the synthetic plant and coupling;
6. a frozen active-POLISH early-warning target, causal arrived-signal feature boundary, disjoint probability/conformal calibration, held-out warning metrics, target sensitivity, and explicit structural/noise failures; and
7. a frozen ten-cause-plus-unknown diagnostic classifier comparison with arrived-only features, mechanistic residuals, compound-event abstention, held-out class metrics, and communication-corruption diagnostics.

2 Evidence taxonomy and claim boundary

The primary measured target is average MRR, denoted $\widehat{\mathrm{MRR}}$. The simulator may derive excursion probability, under-polish or over-polish risk, a hold requirement, and spatial proxies. Without measured spatial metrology, every spatial output is labelled exactly: *Simulated spatial-uniformity proxy; not experimentally validated WIWNU.*

Two evidence planes are kept dimensionally separate. The SI simulator uses Pa, m^3/s, m/s, K, s, and m. The PHM model predicts a stage-average target in the source's native numeric unit because the original source does not declare the target unit and its process columns use hidden scaling factors. An SI physics prediction cannot be added to a residual fitted on scaled PHM inputs without a separately supported calibration bridge.

The supported claims at this evidence freeze are narrow: one selected tree predicts source-native average MRR on the precedence-retained PHM roles; the synthetic CMP subsystem satisfies its declared equations and invariants; configured electrical disturbances can propagate through the tested synthetic utility topology into CMP state; and two frozen models predict one preregistered simulator excursion target on the named primary ensemble. A frozen hybrid also classifies initiating simulator labels conditionally on a valid positive warning and abstains under declared uncertainty gates. The public result has interval undercoverage and temporal fragility; the warning claim rests on only three held-out events and does not survive all audited shifts; the attribution claim uses only six runs per class and loses availability under delay and dropout. The study does not claim prediction or prevention of physical defects, real wafer-yield loss, real equipment damage, controller benefit, or readiness for production control. Feature contributions and rule chains are not described as causal proof.

3 Scientific basis and topology selection

The model begins from Preston-type pressure–velocity proportionality [1] and signed relative-velocity kinematics [2]. Pad asperity wear motivates a stored surface-activity state [3]; conditioning and polishing jointly evolve that state [4,5]; and conditioner geometry or wear motivates a separate dresser-effectiveness state [9,10].

The local literature supports three qualitatively distinct water-related pathways. First, DI water is used during pad conditioning, and conditioning temperature can affect later polishing in the studied apparatus [7,11]. This supports a delayed pathway from an explicit conditioning-water branch to realized dressing, pad state, and later MRR. Second, rinse water can mix with fresh and spent slurry, so a change in rinsing has competing dilution, debris-removal, and slurry-availability effects [8]; its sign is not safely represented by a universal monotone gain. Third, lumped heat storage and slurry heat removal are physically meaningful [6], but the reviewed literature does not establish that a facility UPW header feeds the target tool's coolant loop.

Accordingly, dressing-water support is the primary connected synthetic topology. A thermal loop and a specifically named synthetic slurry-support topology are structural alternatives. No connection is the default and a mandatory negative control. The reviewed papers do not calibrate facility-header pressure or flow thresholds, establish the plumbing of the PHM challenge tool, or justify transferring material-specific numerical effects into this simulator.

4 Data provenance and preprocessing contract

4.1 Archive evidence

The PHM 2016 CMP archive [12] was supplied locally rather than downloaded by the project. The 17,399,074-byte compressed archive has SHA-256 `976b2310...12a1eb9`; all members passed ZIP CRC testing. Selective extraction retained 558 original members: 185 training, 185 test, and 185 validation time-series files plus three MRR tables. Answer files and derived experiments were excluded. Fifty-eight header-only traces remain explicit empty traces; they were not silently imputed or replaced. A separate licence for the embedded dataset was not stated by the source, so redistribution and licensing conclusions are outside this work.

The loader verifies the extraction manifest, file sizes and hashes, exact 25-column time-series headers, three-column MRR headers, identifier presence, duplicate keys, trace counts, and label joins. Targets remain separate from predictors until audited one-to-one joins. Test and validation labels are not joined into fitting features.

4.2 Time support, process proxies, and anomalies

Source row order is preserved and continuity segments break at trace boundaries, non-positive time increments, and gaps exceeding 10 s. Within a continuous segment, the duration represented by row i is

$$w_i = \frac{1}{2}\Delta t^{+}_{i-1} + \frac{1}{2}\Delta t^{+}_{i+1}, \tag{1}$$

where only valid positive adjacent increments contribute. This centered support prevents duplicated timestamps, reversals, or long gaps from receiving fictitious duration. Five process-mode proxies are derived from inputs only; they are not represented as measured phase labels.

The processed bundle contains 1,981 training, 424 test, and 424 validation wafer/stage records, each with 405 target-free predictor columns. All official joins have zero missing and zero orphan keys. Complete-trace features are explicitly offline-only. The audit retains three negative time increments in training, two in validation, 2,907 zero increments, and 2,855 long gaps. It also records 221, 43, and 42 unresolved wafer/stage groups in training, test, and validation, respectively.

Before virtual-metrology fitting, a feature-only identity audit found that the source partitions share 113 training/test, 115 training/validation, and 34 test/validation wafer IDs, always in the opposite stage. We therefore apply fixed source precedence (training, then test, then validation), retaining 1,981, 311, and 275 rows from 1,699, 302, and 267 mutually disjoint wafers. The complete 424-row source holdouts remain collision-contaminated descriptive diagnostics and cannot support independent-wafer generalization claims.

Four training targets lie between 4,129.494 and 4,326.154 while all other training targets are at most approximately 163. Dividing those four values by 60 would place them between 68.825 and 72.103, but no authoritative source supports doing so as a factual correction. The preregistered policy is therefore to retain original labels for the primary analysis, report a sensitivity excluding the four records, and report a separately labelled hypothetical divide-by-60 sensitivity. Selection among these treatments based on holdout performance is forbidden.

Official training is restricted to fitting and tuning; precedence-retained test is an offline holdout; and precedence-retained validation is the final public holdout. Inner resampling retains whole wafers and chronological blocks. A physical-machine holdout is infeasible because all records use machine ID 2; a varying field named `MACHINE_DATA` is not treated as a stable machine identifier.

4.3 Public-data result status

WP09 is complete for offline average MRR in the source-native target scale. The selected tree beats the mean baseline on both precedence-retained holdouts, but its primary split-conformal intervals fall below the frozen coverage floor. The result does not establish physical units, machine generalization, electrical/UPW causality, streaming performance, or control utility.

5 System architecture and causal timing

Figure 1 shows the completed plant and CMP path and the later supervisory stages. At simulation step k, the ordered runtime is: scenario disturbance; electrical and UPS state; VFD and motor; pump; UPW hydraulic and thermal state; utility-to-CMP coupling; true CMP state and MRR; sensor generation; streaming features; prediction; attribution; controller proposal; independent safety filtering; and final action logging.

The sensor contract distinguishes three times:

$$\begin{aligned} t_{\text{source}} &= t_k, \\ t_{\text{reported}} &= \max(0, t_{\text{source}} + \epsilon_{\text{clock}}), \\ t_{\text{arrival}} &= t_{\text{source}} + d_{\text{comm}}. \end{aligned} \tag{2}$$

Reported timestamp jitter does not alter the source or arrival time. At a decision timestamp t_d, an online consumer may use only observations whose $t_{\text{arrival}} \le t_d$, prior predictions and actions, and validated static configuration. It may not use latent truth, future observations, offline excursion labels, or simulator cause labels. Actions proposed at step k take effect no earlier than step $k+1$. Initiating scenario causes are also kept distinct from downstream propagation states such as UPS transfer or VFD derating.

6 Reduced-order utility plant

The utility model is deliberately lumped: it captures causal timing and conservation at the proof-of-concept resolution, not switching waveforms, distributed water hammer, detailed pump performance, or equipment protection.

6.1 Electrical source, UPS, and drive

Electrical scenarios prescribe bounded voltage and frequency events. The UPS has explicit output voltage, output frequency, load, battery energy, transfer, bypass, and recovery states. During battery or transfer operation, accepted discharge follows

$$E_b^{k+1} = E_b^k - \frac{P_L \Delta t}{\eta_{\text{inv}}}. \tag{3}$$

Depletion or overload moves the synthetic model to bypass rather than creating energy. The VFD and motor use bounded commands, derating, slew limits, and first-order speed dynamics. These are engineering approximations and not a model of UPS electrochemistry, switching, protection relays, or motor electromagnetics.

6.2 Pump, hydraulic compliance, and thermal state

Rather than determining flow and head independently from affinity rules, the pump is solved against the network using the speed-scaled quadratic curve

$$\begin{aligned} H_p(Q, n, d) &= d H_{\text{shut}} n^2 - k_Q Q^2, \\ k_Q &= \frac{H_{\text{shut}} - H_{\text{ref}}}{Q_{\text{ref}}^2}, \end{aligned} \tag{4}$$

where n is normalized speed and d is a bounded degradation factor. A check valve excludes reverse flow when network differential pressure exceeds shutoff head. The supply node solves the implicit balance

$$\begin{aligned} \frac{C_h(P_s^{k+1} - P_s^k)}{\Delta t} &= Q_p^k - Q_t(P_s^{k+1}) \\ &\quad - Q_r(P_s^{k+1}) - Q_{\text{rel}}(P_s^{k+1}), \end{aligned} \tag{5}$$

with tool, return, relief, and storage terms and without an after-the-fact pressure clip. Every step audits the discrete mass residual. A well-mixed thermal balance uses actual pump inflow. Water chemistry is not inferred; the model emits only a dimensionless synthetic water-quality deviation index, which has no CMP effect.

For a preregistered 0.70-pu speed case, a 10 ms hydraulic solution differs from a 0.5 ms reference by 36.19 Pa, or 0.0121% of nominal pressure. In a pump trip, pressure falls from 300 to 290.244 kPa on the first step and then decreases monotonically. Generated balance residuals remain below 1.0×10^{-12} m^3/s. These checks establish numerical behavior of the synthetic equations, not manufacturer calibration.

7 Reduced-order CMP subsystem

7.1 Modes and removal accounting

The CMP state machine contains IDLE, PREPARE, POLISH, DRESS, HOLD, RECOVER, and COMPLETE. Let $\chi_p = 1$ only during active polishing and $\chi_d = 1$ only during dressing. Material removal, cumulative active time, and selected wear terms are mode-gated. A forced hold records an explicit reason, and resumption requires 0.5 s of continuous valid recovery dwell in the reference configuration. COMPLETE is terminal until reset.

The model records true instantaneous equivalent MRR R_{true}, cumulative removed thickness H, active polish time t_p, and active-time average MRR:

$$\dot{H} = R_{\text{true}}, \qquad \dot{t}_p = \chi_p, \qquad \overline{R} = \frac{H}{\max(t_p, \varepsilon)}. \tag{6}$$

This accounting is essential for later control evaluation: holding the process cannot appear beneficial merely because instantaneous MRR becomes zero.

7.2 Signed rotary kinematics and exposure

For platen angular velocity ω_p, wafer angular velocity ω_w, center offset r_{cc}, and local wafer polar coordinate (r, θ), vector subtraction gives

$$\begin{aligned} v_R(r,\theta) = \big[&\omega_p^2 r_{cc}^2 + (\omega_p - \omega_w)^2 r^2 \\ &+ 2\omega_p(\omega_p - \omega_w) r_{cc} r \cos\theta\big]^{1/2}. \end{aligned} \tag{7}$$

Angular velocities remain signed. Equal co-rotation produces the constant field $|\omega_p r_{cc}|$, stopped axes produce zero, and counter-rotation produces a nonuniform field. The normalized generalized-Preston exposure is the contact-area mean

$$\Phi_{PV} = \frac{1}{A} \int_A \left(\frac{p(r)}{P_0}\right)^{\alpha} \left(\frac{v_R(r,\theta)}{V_0}\right)^{\beta} \mathrm{d}A. \tag{8}$$

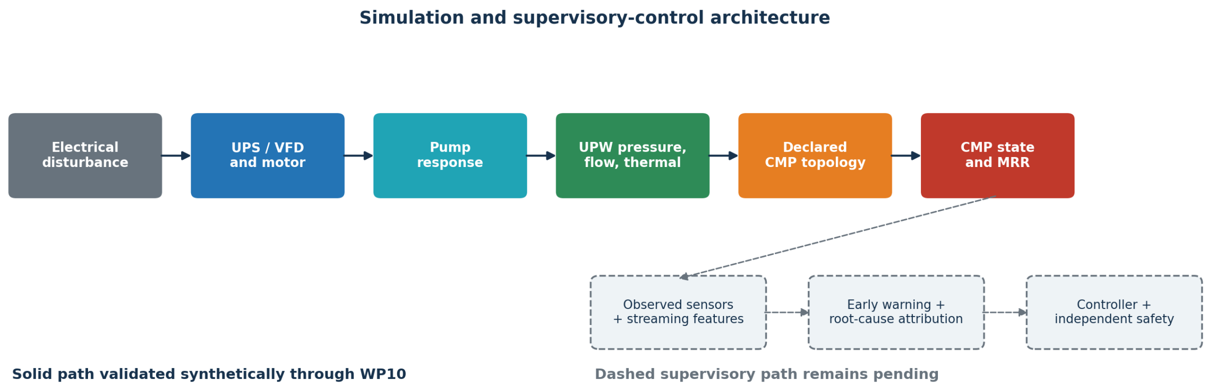

Figure 1: Implemented synthetic propagation path through WP10 and the pending supervisory stages. Solid physical-state propagation is kept separate from observed sensor delivery and from later decision logic. The figure is generated from frozen repository artifacts; predictive-control results are not included.

The integral is evaluated by deterministic radial–angular quadrature. It is not replaced with the product of average pressure and average speed because that equality does not hold in general. The reported annual coefficient of variation is only a *simulated spatial-uniformity proxy; not experimentally validated WIWNU.*

7.3 Consumable memory and dressing

The minimum consumable state comprises reversible pad-surface activity g_p, irreversible remaining pad-life proxy ℓ_p, and dresser effectiveness h_d, each projected inside $[0, 1]$. With dressing command a_d, the frozen model structure is

$$\dot{g}_p = -\chi_p k_g \Phi_{PV} g_p + \chi_d k_c a_d h_d (1 - g_p), \tag{9}$$
$$\dot{\ell}_p = -\chi_p k_{wp} \Phi_{PV} - \chi_d k_{wd} a_d, \tag{10}$$
$$\dot{h}_d = -\chi_d k_d a_d h_d. \tag{11}$$

Dressing can restore reversible surface activity while consuming pad life and dresser effectiveness. This separation prevents a conditioning action from restoring an irreversible life state. The functional decomposition is literature-informed [3–5, 9]; its rate coefficients are synthetic assumptions.

7.4 Slurry availability, interface temperature, and MRR

A bounded dynamic availability state represents the recipe-selected slurry line. The interface temperature T_i is distinct from both slurry and UPW temperature and follows the reduced balance

$$C_i \dot{T}_i = \eta_f \mu A_c P_c \bar{v}_R - U A_T (T_i - T_c) - \rho_s c_{p,s} Q_s (T_i - T_s), \tag{12}$$

where the terms represent frictional input, coolant-boundary removal, and slurry sensible-heat transport. The normalized equivalent removal rate is

$$R_{\text{eq}} = \chi_p R_{0,s} \Phi_{PV} m_s m_T m_{\text{pad}} m_{\text{recipe}}, \tag{13}$$

with dimensionless bounded slurry, temperature, pad, and recipe modifiers. The default material profile fixes the temperature sensitivity at zero. Thus a thermal boundary can change T_i while having exactly zero effect on MRR; a thermal-to-MRR effect requires a separately named non-null material profile.

For the classical $\alpha = \beta = 1$ reference, pressure $P_0 =$ 30,000 Pa, platen/head speeds 8/6 rad/s, and power-mean speed $V_0 = 1.6070416183328249$ m/s, the reference removal is

$$R_0 = 1.6666666666667 \times 10^{-9}\ \text{m/s} = 100\ \text{nm/min}. \tag{14}$$

The corresponding derived Preston coefficient is $K_P = 3.4570078908840606 \times 10^{-14}\ \text{Pa}^{-1}$, satisfying $K_P P_0 V_0 = R_0$ to floating-point tolerance. The numerical values define the synthetic reference recipe; they are not calibrated to the PHM target or a real material stack.

8 Typed utility-to-CMP coupling

The coupler is stateless and consumes only latent UPW state. Its output is an already defined typed CMP boundary. It cannot inspect raw voltage, UPS, VFD, pump, or corrupted sensor fields; calculate MRR; inject a generic discrepancy multiplier; force a hold; or take a control action.

For normalized pressure $p = P_s/P_{\text{ref}}$ and tool flow $q = Q_t/Q_{\text{ref}}$, define the clipped ramp

$$\mathcal{R}(x; x_0, x_1) = \text{clip}\left(\frac{x - x_0}{x_1 - x_0}, 0, 1\right). \tag{15}$$

Hydraulic support and link-adjusted availability are

$$\begin{aligned} a_h &= \min\{\mathcal{R}(p; p_0, p_1), \mathcal{R}(q; q_0, q_1)\}, \\ a_{\text{eff}} &= 1 - \lambda(1 - a_h). \end{aligned} \tag{16}$$

The minimum treats pressure and flow as correlated indicators of the same hydraulic network and avoids multiplying the disturbance twice. The ramp, minimum, and affine blend are engineering approximations; every numerical threshold, reference, and link strength is a synthetic assumption.

The four frozen structures are:

No connection: exact neutral boundary and mandatory default.

Dressing-water support: a_{eff} changes realized dressing availability only, allowing a delayed effect through Eq. (9).

Thermal loop: cooling conductance is scaled by a_{eff} and

$$T_c = T_{c,0} + \lambda(T_u - T_{u,\text{ref}}). \tag{17}$$

Neutral CMP and upstream temperatures are distinct, making zero link exactly neutral for any UPW temperature and nominal UPW temperature neutral for any link strength.

Synthetic slurry support: a_{eff} changes only a specifically named synthetic slurry-utility availability boundary.

Nominal connected references are 300 kPa supply pressure and 100 mL/s tool flow. Pressure zero/full fractions are 0.60/0.95, while flow zero/full fractions are 0.50/0.95. These define the synthetic experiment and are not operating limits for any real CMP tool. Every coupling parameter has a unit, hard bound, sensitivity range, expected direction, and provenance class.

9 Validation design

The scientific checks cover dimensions, limiting cases, monotonicity, state bounds, process-mode gating, energy and mass residuals, deterministic replay, quadrature convergence, timestep refinement, null topologies, and parameter mismatch. Two full-chain cases serve different purposes.

The negative control is a 25% voltage sag lasting 400 ms with a healthy UPS. It asks whether the coupling remains inactive when the synthetic utility plant rides through the event. The positive chain is separately configured as a degraded 500 J UPS with 2,500 W load, 0.95 inverter efficiency, a grid interruption from 1 to 4 s during a 6 s dressing interval, and an explicit VFD restart request. A 1 s preparation interval follows and polishing starts at 7 s. The connected and no-connection runs share identical upstream states, initial conditions, timestep, and disturbance.

Local derivatives are evaluated only away from ramp and minimum knots. Global sensitivity uses seeded Saltelli pick–freeze Monte Carlo estimators [13]: 4,096 base samples, seven independent parameter ranges, seed 20260711, and 36,864 evaluations at one representative degraded state. Raw finite-sample indices are retained without clipping or renormalization. Structural and mismatch cases include no connection, zero link, several link strengths, and earlier- and later-response thresholds.

10 Results through WP10

All numerical results in this section are **Synthetic-simulation evidence** and conditional on the specified configuration. The public-data archive evidence in Section 4 is provenance and schema evidence, not a public-model performance result.

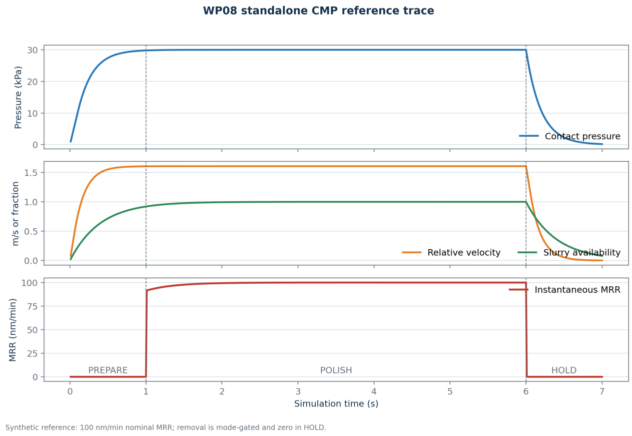

Figure 2: Frozen WP08 synthetic reference trace. Material removal is gated to POLISH; cumulative removal and active-time average retain process history across non-polishing modes.

10.1 Standalone CMP invariants and reference trace

The 16×64 radial–angular quadrature gives $V_0 = 1.6070416183328249$ m/s. Nominal exposure differs from one only at floating-point precision, and both zero pressure and zero speed produce exactly zero exposure and removal. Across 8×32, 16×64, 32×128, and 48×192 grids, the constant-field integral differs from one by at most 2.22×10^{-16} and nominal area-mean velocity differs from the finest grid by at most 4.44×10^{-16} m/s.

The nominal annular exposure coefficient of variation is 0.0025316618572622336. This is a *simulated spatial-uniformity proxy; not experimentally validated WIWNU*.

Figure 2 contains 1 s PREPARE, 5 s POLISH, and 1 s HOLD at a 10 ms step. Only polishing increases active time and cumulative removal. Final active time is 4.999999999999938 s, cumulative removal is $8.276153498157771 \times 10^{-9}$ m, and active-time average MRR is $1.6552306996315749 \times 10^{-9}$ m/s. The average is below the fresh-pad 100 nm/min reference because pad activity decays during polishing.

A separate 10 s dressing limit starts at pad activity 0.5, remaining pad life 0.8, and dresser effectiveness 0.9. It ends at 0.5823912561, 0.7995, and 0.8998200179, respectively, with exactly zero instantaneous MRR. The case confirms restoration of reversible surface activity without restoration of remaining life.

The maximum absolute discrete interface-energy residual over the 700-step reference is $1.4188941577231162 \times 10^{-8}$ W, below the 10^{-7} W audit tolerance. With default thermal sensitivity zero, cold and warm coolant boundaries produce different temperatures after 1 s but exactly zero instantaneous-MRR difference.

Table 1 reports refinement of a 70% pressure-command case against a 1 ms reference. All three error measures decrease monotonically, supporting the 10 ms default within the tested standalone configuration.

Table 1: Standalone CMP timestep refinement.

Δt (s)	$\lvert e_P \rvert$ (Pa)	$\lvert e_T \rvert$ (K)	$\lvert e_H \rvert$ (m)
0.0400	25.8847	2.8404×10^{-4}	1.9141×10^{-11}
0.0200	13.5017	1.3801×10^{-4}	9.3021×10^{-12}
0.0100	6.6010	6.5280×10^{-5}	4.4005×10^{-12}
0.0050	2.9787	2.8992×10^{-5}	1.9545×10^{-12}
0.0025	1.1254	1.0868×10^{-5}	7.3269×10^{-13}

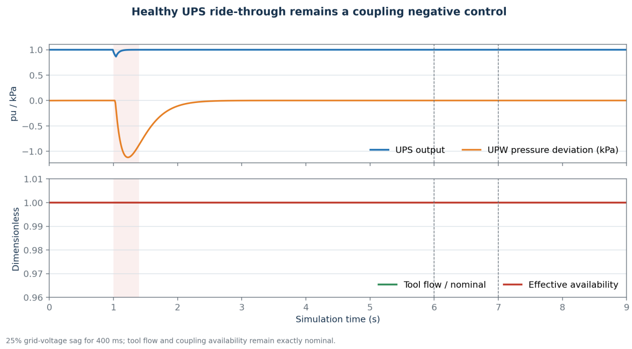

Figure 3: Healthy-UPS synthetic negative control. The electrical sag changes upstream states slightly, but the declared dressing-water connection remains at full support and connected/no-connection CMP traces are identical.

At the WP08 evidence freeze, 48 focused CMP/contract checks and all 130 repository tests passed. This test evidence supports the declared synthetic structure and numerics only.

10.2 Healthy-sag negative control

The healthy-UPS 25%, 400 ms sag is replayed after 3 s plant warm-up, followed by 6 s of dressing, 1 s preparation, and polishing from 7 s. Minimum UPS output is 0.865536 pu; minimum motor speed is 187.1656743 rad/s; minimum pump flow is 1.9724375×10^{-4} m^3/s; and minimum supply pressure is 298.8797567 kPa. Tool flow remains 100 mL/s. Both hydraulic and effective support remain exactly one.

As shown in Fig. 3, connected and no-connection upstream traces are identical by construction, and their pad activity, MRR, and cumulative-removal traces also match exactly. The connected trace SHA-256 is `f8359518...bfc3f20`. This result is a structural negative control, not a failed attempt to force an excursion: the coupling was not tuned to turn a well-ridden-through event into a CMP perturbation.

10.3 Declared positive causal chain

The positive synthetic stress case produces the ordered onsets in Table 2. The minimum UPS output approaches zero, pump flow reaches zero, supply pressure approaches the 100 kPa return boundary, tool flow approaches zero, and dressing availability reaches zero. Upstream traces remain identical between the connected and no-connection variants.

Table 2: First source-time onsets in the positive synthetic chain.

Event	Time (s)
Grid interruption	1.00
UPS output < 0.9 pu	1.03
Pump flow $< 99\%$ reference	1.04
Motor speed $< 99\%$ nominal	1.19
UPW pressure below full support	1.24
Effective availability < 1	1.24
Polishing starts	7.00

The connected boundary affects only dressing availability. Therefore MRR is exactly zero during dressing in both variants. By the end of dressing, pad activity is 0.5511876609 with no connection and 0.5216572820 with dressing-water support, a 5.3576% relative decrease. When polishing begins after utility recovery, the stored pad difference changes later removal (Table 3 and Fig. 5).

The positive no-connection and connected trace SHA-256 values are `3560cb28...41fc3f20` and `19f440b2...51dfa10`, respectively. The delayed 3.1926% effect is conditional on the selected synthetic topology and values; it is neither a safe-envelope violation nor evidence of controller benefit.

10.4 Sensitivity and structural uncertainty

Off-knot finite-difference signs match the frozen expectations. Increasing link strength, reference pressure or flow, or the transition thresholds reduces availability at their selected transition states. In the 0.5-link thermal case, increasing the UPW reference temperature changes mapped coolant temperature with derivative -0.5. Derivatives at ramp and minimum knots are deliberately not reported.

Figure 6 and Table 4 show that link strength and reference pressure dominate this selected analysis. A small first-/total-order inconsistency for the flow-full fraction is preserved as a finite-sample estimator result, rather than corrected cosmetically. Across zero, 0.25, and 1.0 link strengths and alternate threshold sets, mean later MRR spans $9.8585309520 \times 10^{-10}$ to $1.0198314108 \times 10^{-9}$ m/s. Zero link exactly matches no connection. The simulated effect is therefore materially contingent on structure and parameter assumptions.

At the WP10 evidence freeze, all 50 coupling-impacted tests and all 154 repository tests passed. The runtime schema/interface versions are 2.4.0/3.2.0 and the default topology remains no connection with link strength zero. Test counts certify the checked software contracts and synthetic invariants, not real-fab validity.

11 Virtual metrology, early warning, and attribution

WP09 is a frozen offline public-data result, WP12 a frozen synthetic early-warning result, and WP13 a conditional synthetic attribution result. Control, safety,

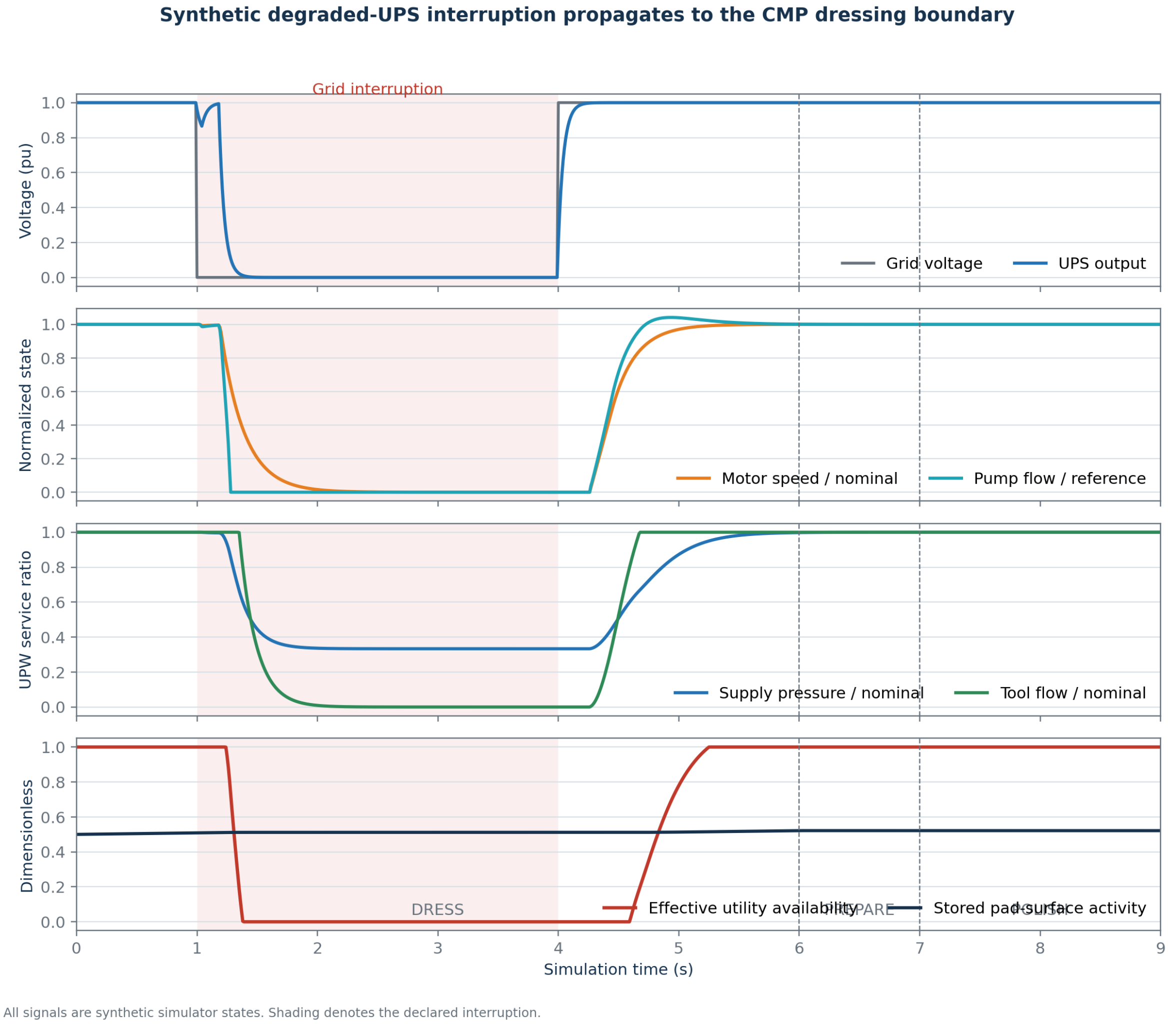

Figure 4: Positive synthetic full-chain stress test. A deliberately degraded UPS during DRESS propagates through the drive, pump, and UPW state to dressing availability. This experiment is not a real UPS specification or real-tool plumbing model.

and integrated-runtime subsections remain protocols. The WP12/WP13 runners are open loop: they do not propose or apply controller actions and are not the WP17 runtime.

11.1 WP09: public-data virtual metrology

The one-shot WP09 study compares a training-target mean, ordinary linear and ridge regression, a dimensionless native-scale physics proxy, histogram gradient boosting, and a physics-plus-residual tree. All scientific choices, the 405-column predictor order, whole-wafer roles, and interpretation gates were committed before official holdout targets were opened. Training wafers were assigned by feature identity to TRAIN_CORE (1,396 rows/1,189 wafers), MODEL_SELECTION (299/255), and INTERVAL_CALIBRATION (286/255). Five grouped folds inside TRAIN_CORE tune hyperparameters; MODEL_SELECTION recommends a family; final development fitting combines the first two roles; and the third supplies interval scores. The retained test/final-validation roles contain 311/275 rows from 302/267 additional whole wafers.

For six active-polish proxy pressures, fit-only positive medians m_j^+ give $P_i^* = \frac{1}{6}\sum_j \max(0, p_{ij})/m_j^+$. Absolute wafer, stage, and head rotations are similarly normalized, with $V_i^* = (\max(r_{w,i}^*, r_{s,i}^*) + r_{h,i}^*)/2$. The declared exposure and native coefficient are

$$\phi_i = \mathbb{I}[\text{active support}] P_i^* V_i^*,$$
$$\widehat{K}_{native} = \max\left(0, \frac{\sum_i \phi_i y_i}{\sum_i \phi_i^2}\right). \tag{18}$$

This is not an SI Preston coefficient. The hybrid predicts $\max(0, \widehat{K}_{native}\phi_i + f_\theta(X_i))$. Common fit-only median imputation appends one missing indicator per predictor and removes zero-variance derived columns. Linear/ridge designs are standardized; tree designs are not. No target-driven feature selection occurs.

For calibration residuals $s_i = |y_i - \widehat{y}_i|$, the symmetric 90% split-conformal radius uses

$$k = \min\{n, \lceil (n+1)(1-0.10) \rceil\},$$
$$q = s_{(k)}, \tag{19}$$
$$[L_i, U_i] = [\max(0, \widehat{y}_i - q), \widehat{y}_i + q].$$

The selected tree uses absolute loss, 15 leaves, minimum

Table 3: Positive synthetic chain result. Values are paired against identical upstream states; no safe envelope or controller is involved.

Metric	No connection	Dressing-water support	Relative decrease
End-of-dress pad activity	0.5511876609	0.5216572820	5.3576%
Mean later-polish MRR (m/s)	$1.0198314108 \times 10^{-9}$	$9.8727270376 \times 10^{-10}$	3.1926%
Final cumulative removal (m)	$2.0396628215 \times 10^{-9}$	$1.9745454075 \times 10^{-9}$	3.1926%

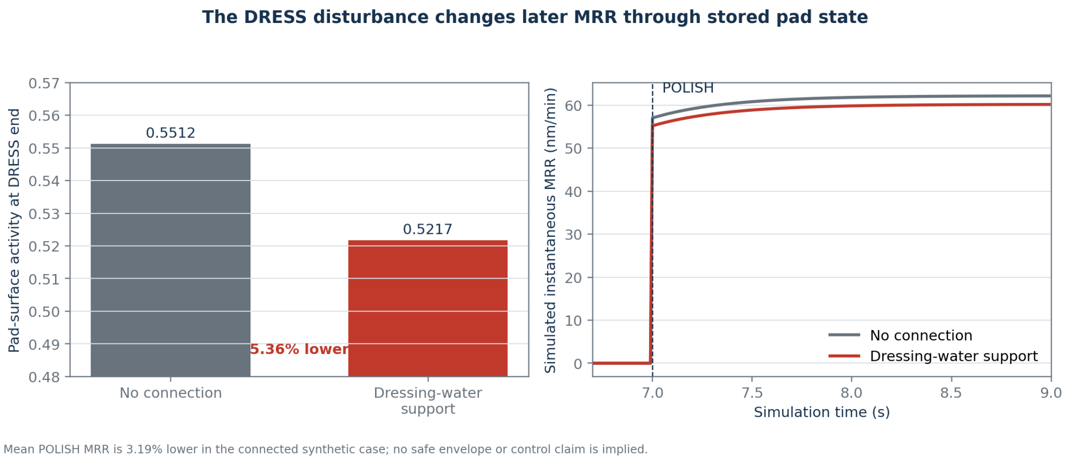

Figure 5: Delayed synthetic CMP result. Utility support changes pad activity during DRESS; only later, during POLISH, does MRR differ. There is no direct instantaneous UPW-to-MRR multiplier.

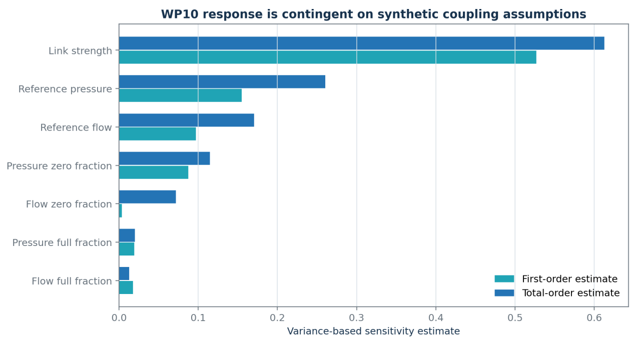

Figure 6: Raw Saltelli first- and total-order estimates for effective availability at one representative degraded state. These are conditional simulator sensitivities, not empirical feature importance or causal proof.

leaf size 20, L2 regularization 0.1, and radius $q = 5.3498$ in the source-native target scale.

The tree was recommended on MODEL_SELECTION before holdout access and reduces MAE relative to the mean by 89.22%/88.57%. The final-validation whole-wafer bootstrap interval for paired tree-minus-mean MAE is $[-28.191, -24.363]$, so the preregistered public-prediction gate passes. Its test/final-validation grouped-bootstrap MAE intervals are $[2.779, 3.737]$ and $[2.784, 4.108]$. Relative MAE is 3.572%/3.858%.

The hybrid is worse than the tree on both roles; the final-validation paired hybrid-minus-tree interval is $[1.502, 3.650]$. Thus the physics-plus-residual improvement claim fails. Ordinary least squares has severe extrapolation and 22/23 clipped negative predictions on test/validation. The physics proxy is also worse than the mean in MAE; it remains a structural baseline rather than a calibrated physical model.

The primary tree intervals fail the frozen 0.85 coverage floor on both roles. Test Stage A/B coverage is 0.853/0.818 and validation coverage 0.852/0.840, so the shortfall is not isolated to one role. Gap-threshold (5/20 s), active-only, and no-unresolved-proxy sensitivities keep tree MAE near 3.0–3.6 and generally increase coverage, but secondary variants cannot rescue the primary gate. Excluding the four preregistered extreme training labels yields MAE 3.050/3.100; the separately labelled hypothetical divide-by-60 sensitivity yields 3.032/3.356. The original-label result remains primary.

Consumable-feature removal worsens tree MAE but improves ridge MAE, so the cross-model association gate fails and no causal degradation claim is made. Chronological stress is also severe: tree MAE/RMSE become 18.50/239.84 with $R^2 = 0.037$ and coverage 0.812. Median absolute error remains 3.19, but one preregistered extreme late-block label (target 4326.154 versus prediction 152.927 and late-block median target 83.309) dominates the tail. No post-hoc exclusion model was fitted. Machine generalization remains infeasible because all records expose the same machine identifier.

11.2 WP12: early-warning target and models

Let R_k be disturbed true simulated instantaneous average MRR and R_k^{ref} the event-disabled paired reference at the same step. The reference preserves initial state, schedule, declared topology, and static plant parameters.

Table 4: Seeded global sensitivity of effective availability. Raw estimates are conditional on the chosen state, ranges, topology, and output.

Parameter	First order	Total order	Sampled range
Link strength	0.5268	0.6127	$[0, 1]$
Reference supply pressure (Pa)	0.1551	0.2603	$[250{,}000, 350{,}000]$
Pressure zero fraction	0.0875	0.1147	$[0.40, 0.75]$
Pressure full fraction	0.0194	0.0201	$[0.85, 1.00]$
Reference tool flow (m^3/s)	0.0971	0.1705	$[8 \times 10^{-5}, 1.2 \times 10^{-4}]$
Flow zero fraction	0.0038	0.0718	$[0.30, 0.70]$
Flow full fraction	0.0178	0.0128	$[0.85, 1.00]$

Table 5: WP09 primary precedence-retained holdout results. Target values are source-native and the unit is undeclared. Coverage is empirical for the nominal 90% split-conformal interval.

Family	Test MAE	Val. MAE	Test RMSE	Val. RMSE	Test R^2	Val. R^2	Test/val. coverage
Mean	29.540	29.703	33.152	33.170	-0.022	-0.020	0.852/0.865
Linear	101.255	215.637	413.773	1955.543	-158.132	-3542.915	0.894/0.895
Ridge	28.426	35.325	47.449	65.038	-1.093	-2.920	0.913/0.884
Physics proxy	34.077	35.709	61.086	62.124	-2.468	-2.577	0.875/0.887
Tree	**3.185**	**3.395**	**5.176**	**6.693**	**0.975**	**0.958**	**0.839/0.847**
Hybrid	4.723	5.935	8.251	11.704	0.937	0.873	0.916/0.905

The frozen point-violation indicator is

$$v_k = \mathbb{I}[m_k = \textsc{polish}]\mathbb{I}[R_k^{\mathrm{ref}} > 10^{-12}\,\mathrm{m/s}] \\ \times \mathbb{I}[R_k < 0.95R_k^{\mathrm{ref}} \ \vee \ R_k > 1.05R_k^{\mathrm{ref}}]. \tag{20}$$

An episode requires $n_p = \lceil 0.25\,\mathrm{s}/0.01\,\mathrm{s} \rceil = 25$ consecutive active-POLISH violating intervals. Its onset is the first interval of the subsequently persistent departure. Other modes cannot create an event merely because their MRR is zero.

At eligible 0.10 s decision time t_d, with $H = 3.0$ s, the offline label is

$$y(t_d) = \mathbb{I}[t_d < t_e \le t_d + H], \tag{21}$$

where t_e is the first accepted onset. Rows are censored instead of made negative if the full horizon is absent, no 0.25 s contiguous active-POLISH opportunity exists, or onset has already occurred.

Only observations satisfying $t_{\mathrm{arrival}} \le t_d$ and $k_{\mathrm{source}} \le k_d$ enter the 65-feature vector. Eight observed channels cover grid voltage, UPS output/battery, motor speed, pump flow, UPW pressure/tool flow, and UPW temperature. Each contributes latest value and age, 0.5 s mean/minimum/slope/missing fraction, and history minimum. UPS output, pressure, and tool flow also use arrived-history deficit

$$D_j(t_d) = \int_0^{t_d} \max(0, 0.95 - z_j(t))\,dt. \tag{22}$$

Known synthetic schedule indicators and static battery/load ratios are permitted. MRR, latent pad/coupling state, event family or cause, and future truth are prohibited. Imputation and scaling fit TRAIN only.

Whole runs are disjoint across TRAIN (2,988 eligible/84 positive rows), probability CALIBRATION (1,404/84), independent CONFORMAL_CALIBRATION (1,404/84), and TEST (1,932/84). The 84 positive TEST rows are 28 warning decisions around each of only three event-bearing runs. The frozen comparison uses a training-prevalence constant, balanced logistic regression, and balanced histogram gradient boosting. No TEST metric tunes a model or selects one for sensitivity/robustness analysis.

With calibrated probability $\tilde{p}_i$, independent conformal rows use

$$s_i = \begin{cases} 1 - \tilde{p}_i, & y_i = 1, \\ \tilde{p}_i, & y_i = 0, \end{cases} \tag{23}$$

and the prediction set is

$$\Gamma(x) = \{0 : \tilde{p}(x) \le \hat{q}\} \\ \cup \{1 : 1 - \tilde{p}(x) \le \hat{q}\}. \tag{24}$$

where $\hat{q}$ is the finite-sample higher quantile at $\alpha = 0.10$. Temporal dependence and fixed severity grids make coverage an empirical row-level simulator diagnostic, not a run-wise or real-world guarantee.

Both learned models pass the four frozen primary gates, but the gate has no false-alarm acceptance threshold and event recall rests on three events. The structural-null set has zero true events, yet logistic regression raises 30 false-alarm episodes (specificity 0.3371) and gradient boosting nine (specificity 0.2860). Under high synthetic noise, logistic PR-AUC/precision/ specificity/coverage become 0.6002/0.1019/0.0657/0.0959; gradient boosting gives 0.5283/0.5283/0.9053/0.7671. These failures show that upstream severity can be learned without proving downstream CMP applicability. WP15/WP16 must therefore gate topology, sensor validity, uncertainty, and model envelope.

11.3 WP13: root-cause attribution

The ten initiating classes are voltage sag, voltage swell, interruption, frequency deviation, pump trip, valve restriction, demand spike, thermal excursion, pressure-sensor fault, and flow-sensor fault. UNKNOWN is a required eleventh output. UPS transfer and VFD derating are recorded only as propagation evidence. Simulator

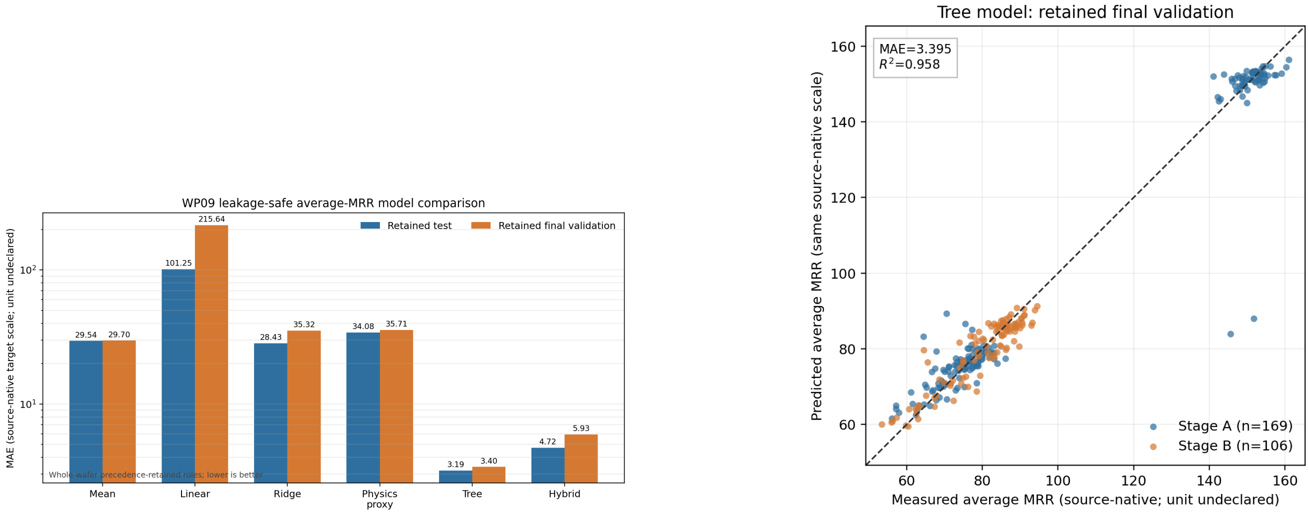

Figure 7: WP09 public-data results in the source-native target scale. Left: all six frozen model families on mutually whole-wafer-disjoint retained roles; the MAE axis is logarithmic. Right: tree predictions on retained final validation, stratified by source stage. Neither axis has a declared physical unit.

Table 6: Primary held-out WP12 warning results. One false-alarm episode over 193.2 eligible simulated seconds is 18.63 per eligible simulated hour; this is not a real-fab alarm rate.

Metric	Prevalence	Logistic	Gradient boosted
PR-AUC (prevalence 0.04348)	0.04348	0.99040	0.91304
Precision	undefined	0.97500	0.91304
Row recall	0.00000	0.92857	1.00000
Specificity	1.00000	0.99892	0.99567
Brier score	0.04186	0.00321	0.00678
Event recall	0/3	3/3	3/3
Median warning lead (s)	undefined	2.71	2.71
False-alarm episodes	0	1	1
Conformal coverage	0.95652	0.92961	0.87733
Empty-set fraction	0.00000	0.07039	0.12267

labels are joined after estimation for offline scoring and are absent from the online window

$$\mathcal{O}(t_d) = \{o_i : t_{\mathrm{arrival},i} \le t_d\}. \tag{25}$$

The estimator computes latest value, 0.75 s mean, least-squares slope, and missing fraction for 14 observed utility and automation signals. It excludes future records, latent state, scenario metadata, CMP MRR, pad and coupling state, and offline excursion or cause labels.

Pressure and flow consistency evidence is

$$\begin{aligned} \widehat{P} &= P_r + (P_0 - P_r)(Q_p/Q_{p,0})^2, \\ r_P &= (P_{\mathrm{obs}} - \widehat{P})/P_0, \\ \widehat{Q}_t &= \min[D, v\max(P_{\mathrm{obs}} - P_r, 0)/R_t], \\ r_Q &= (Q_{\mathrm{obs}} - \widehat{Q}_t)/Q_{t,0}. \end{aligned} \tag{26}$$

These are engineering diagnostic proxies, not independent hydraulic truth. Sensor-fault scenarios corrupt observations without changing latent hydraulics. Rule severities use

$$\rho(d; a, b) = \mathrm{clip}\left((d - a)/(b - a), 0, 1\right). \tag{27}$$

A balanced multinomial logistic comparator uses TRAIN-only median imputation and standardization and a probability temperature selected on disjoint CALIBRATION runs. The signed contribution $a_{cj} = \beta_{cj}\widetilde{x}_j$ explains a fitted logit only; it is not a causal effect. The preregistered primary method pools rule and learned probabilities geometrically:

$$\begin{aligned} s_c &= 0.5\log(p_{\mathrm{ML},c} + 10^{-9}) \\ &\quad + 0.5\log(p_{\mathrm{rule},c} + 10^{-9}), \\ p_{\mathrm{hyb}} &= \mathrm{softmax}(s). \end{aligned} \tag{28}$$

Negative or uncertain warnings, stale or missing sensors, out-of-distribution features, low confidence or margin, rule/model conflict, and multiple strong initiating chains force UNKNOWN. The primary diagnostic study supplies a neutral valid-positive singleton warning to isolate cause separability from WP12 recall; it is not an end-to-end operational rate.

TEST contains 66 disjoint whole runs: six for each known cause and six normal UNKNOWN cases. Table 7 and Fig. 10 report the frozen result. The hybrid correctly classifies 57/66 runs. All nine errors are conservative abstentions; it never substitutes an incorrect known cause. Pressure-sensor-fault recall is weakest at 0.3333. The rule-only comparator scores higher than the primary hybrid, but post-TEST model switching is prohibited.

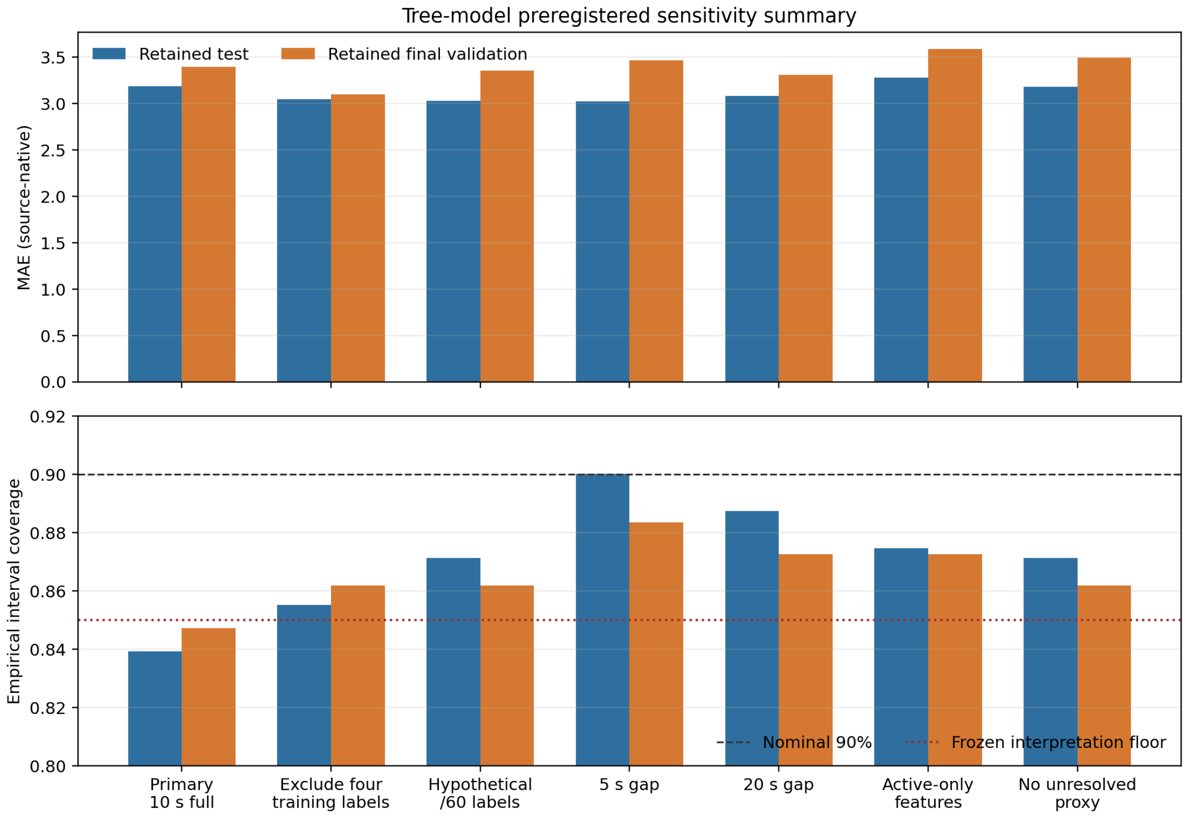

Figure 8: Preregistered tree sensitivities. Point MAE is stable across declared label, continuity, and feature-proxy variants, whereas empirical interval coverage changes materially. The dashed line is nominal 90% and the dotted line is the frozen 85% interpretation floor.

Table 7: Held-out WP13 whole-run attribution. Coverage is the fraction of known-cause runs receiving a known label; selective accuracy conditions on such labels. All quantities are conditional synthetic diagnostic results.

Method	Accuracy	Macro recall	Unknown recall	Known coverage	Selective accuracy	Top-2
Always unknown	0.0909	0.0909	1.0000	0.0000	–	0.1818
Rule only	0.9242	0.9242	1.0000	0.9167	1.0000	1.0000
Logistic	0.7121	0.7121	0.8333	0.7167	0.9545	0.8182
Hybrid (primary)	0.8636	0.8636	1.0000	0.8500	1.0000	1.0000

Grouped-bootstrap hybrid accuracy has 5th–95th percentiles 0.7879–0.9242. On six unseen interruption/demand compounds, the estimator abstains on 5/6 and its pre-abstention top two contain both offline truth labels on 6/6. Doubled noise and parameter mismatch retain 0.8636 aggregate accuracy. In contrast, 0.20 s delay forces all runs to UNKNOWN, and 10% packet loss reduces known-cause coverage to 0.15 while selective accuracy remains 1.0. These are diagnostic-availability failures rather than false known-cause substitutions. Hybrid local inference has median/95th-percentile times of 3.423/3.746 ms; these are not production-latency guarantees.

11.4 WP14–WP16: baselines, predictive supervision, and safety

PENDING: baselines will include no action, fixed threshold with hysteresis, safe hold, and controlled resume. The first predictive supervisor will use bounded action search or short-horizon model-predictive control, not reinforcement learning. Candidate actions are no action, bounded VFD or valve adjustment, bounded downforce/head/platen-speed reduction, advisory warning, safe hold, and controlled resume.

The optimization objective must jointly penalize predicted active-polish MRR deviation, cumulative-removal error, action effort and slew, hold duration, recovery time, and throughput cost. Coefficients, horizon, terminal treatment, and infeasibility behavior remain to be frozen. Cumulative removal and active time from Eq. (6) prevent a hold-only policy from gaming instantaneous-MRR error.

An independent safety filter will evaluate the controller proposal after the controller acts. It will check action magnitude, slew rate, sensor validity, prediction uncertainty, approved simulation envelope, and restart conditions. Its auditable outcomes are APPROVED, CLIPPED, REJECTED-OUT-OF-ENVELOPE, REJECTED-SENSOR-INVALID, REJECTED-HIGH-UNCERTAINTY, and REPLACED-WITH-HOLD. A claim that final simulated actions remain within configured limits requires exhaustive

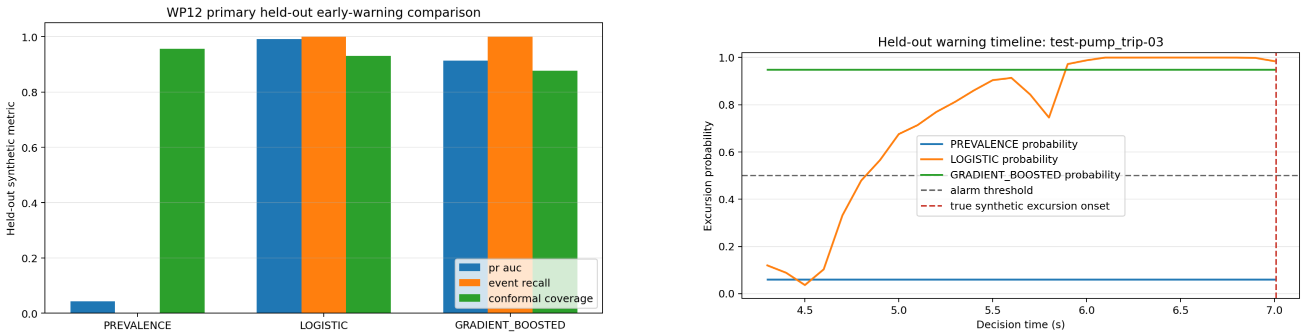

Figure 9: Frozen WP12 synthetic results. Left: held-out discrimination, event accounting, and empirical conformal summaries. Right: all three frozen probability trajectories on one event-bearing held-out run. These are open-loop simulator results, not a control comparison.

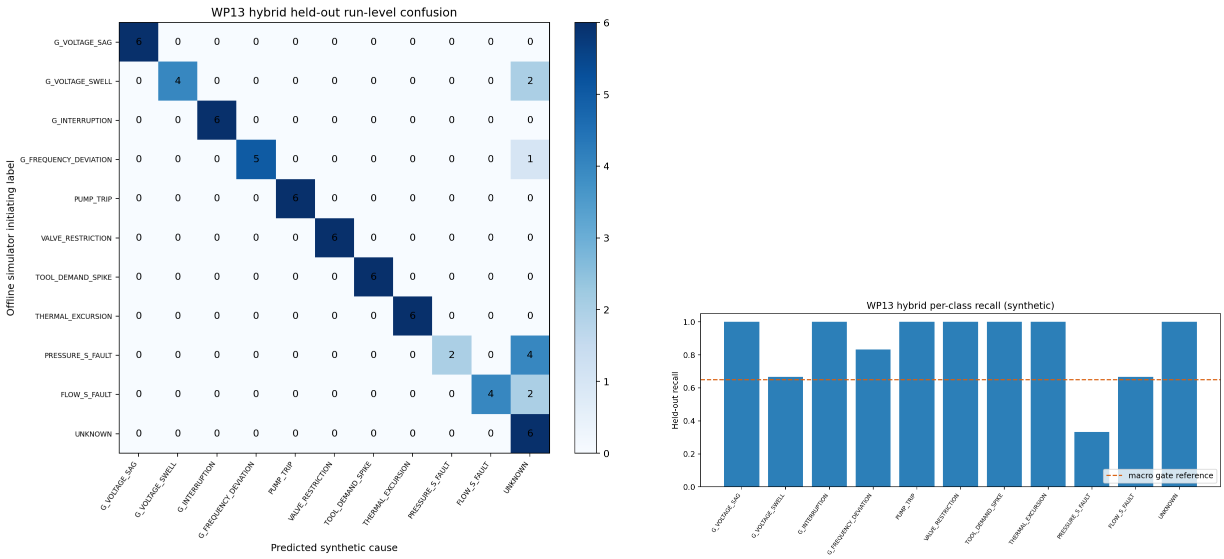

Figure 10: Frozen WP13 hybrid result. Left: run-level confusion matrix; every off-diagonal count is an abstention to UNKNOWN. Right: held-out recall by initiating simulator label. These are synthetic classifications, not experimental causal identification.

rejection-path and property tests with zero final-action violations. It would be a software/simulation property, not equipment certification.

11.5 WP17–WP18: integrated paired evaluation

PENDING: no action, threshold control, and predictive control will be run on identical scenarios, initial states, parameter draws, seed maps, and sensor corruptions. A manifest equality assertion will differ only in controller identity and the trajectories subsequently caused by its actions. The final comparison will include normal, mild and severe electrical disturbances, UPS transfer, pump trip, valve restriction, demand spike, temperature event, sensor bias/dropout, recoverable and hold-required disturbances, slow drift, and compound unseen cases.

Metrics will include peak and integrated active-polish MRR deviation, cumulative-removal error, pressure/flow violation duration, warning time, false alarms, missed events, hold and recovery duration, action magnitude, safety-filter rejections, attribution accuracy, prediction/controller latency, and simulation throughput. Summaries will report mean, median, standard deviation, 5th and 95th percentiles, and worst case. A controller-benefit claim also requires preregistered paired success and non-inferiority criteria; no such claim is made here.

12 Discussion

The most important modelling result is not the magnitude of the positive case but the mechanism and its controls. A direct UPW-to-MRR multiplier would make MRR respond even outside active polishing, hide whether utility water actually connects to the tool process, and allow a gain to manufacture the desired causal chain. In the present topology, electrical loss first depletes a synthetic UPS, then changes drive, pump, and conserved UPW states. Only a declared dressing branch changes CMP boundary availability; the pad activity state stores that history; and MRR changes only during subsequent polishing. The healthy-sag null case and ex-

Table 8: Manuscript evidence status at the WP09/WP12/WP13 freeze. "Protocol" means the method is described but no performance result is asserted.

Work package	Topic	Status	Current manuscript treatment
WP03/R2	PHM data pipeline	Verified contract	Provenance, semantics, native-unit and split policy
WP04–WP07/R3–R4	Utility, sensing, timing	Synthetic checks complete	Equations, conservation, causal observation contract
WP08	CMP physics	Synthetic checks complete	Reference, limiting, energy, timestep, replay results
WP10	Utility–CMP coupling	Synthetic checks complete	Null/positive chain and sensitivity results
WP09	Virtual metrology	Public-data result	Six-family comparison, grouped uncertainty, sensitivities and failed gates
WP12	Early warning	Synthetic result	Frozen target, causal features, disjoint calibration, primary and shift results
WP13	Attribution	Synthetic result	Arrived-only evidence, abstention, whole-run class and robustness results
WP14–WP16	Control and safety filter	Protocol	Bounded actions, objectives, independent validation plan
WP17–WP18	Integrated comparison	Protocol	Paired-run and robustness plan

act zero-link equivalence are as important as the positive chain because they constrain model freedom.

The 3.19% decrease is a within-simulator paired difference, not an estimate of what any real CMP tool would experience. Global sensitivity assigns substantial conditional influence to link strength and reference pressure, while mismatch cases span the no-connection result. This means topology and coupling uncertainty must be propagated into later warning and controller studies. A predictive policy that succeeds only when an assumed link is strong may fail under no connection or a different plumbing path; later evaluation must retain those structural nulls rather than optimize them away.

WP12 makes that warning concrete but also demonstrates the applicability problem. Primary held-out separation is high and both learned models warn of all three configured events 2.71 s in advance at the median. However, the severe anchors repeat a narrow full-DRESS service-loss mechanism, so 84 positive decision rows do not constitute 84 independent events. Whole-run bootstrap intervals resample only the same 24 configured TEST runs and cannot create missing physical diversity.

More importantly, the no-connection diagnostic breaks the implication from upstream severity to downstream MRR: the models alarm frequently while the primary MRR target remains negative. High noise also produces poor discrimination, false alarms, and empty conformal sets. A prediction score is therefore neither root-cause proof nor permission to actuate. This negative result directly constrains WP15/WP16: applicability and uncertainty failures must reject the proposal or replace it with a safe hold, independently of nominal TEST accuracy.

WP13 adds a second distinction: diagnostic correctness versus diagnostic availability. On the primary grid the hybrid never replaces one known cause with another, but it abstains on nine known-cause runs. This produces perfect selective accuracy at 0.85 known-cause coverage. Pressure-sensor faults expose the limited independent redundancy of the reduced-order plant, while delay and dropout drive the estimator toward UNKNOWN. A runtime must not treat UNKNOWN as evidence that the process is normal. The rule-only comparator also exceeds the hybrid on held-out accuracy, showing that learned fusion has not demonstrated incremental value in this ensemble. Preserving the frozen primary avoids post-hoc selection; it does not hide the stronger comparator. Rule chains, residual agreement, and coefficient contributions remain diagnostic explanations within the simulator rather than causal proof.

The explicit evidence-plane separation is equally consequential for virtual metrology. Public PHM features are scaled and the target unit is not declared by the original source. The simulator can still supply hypotheses and dimensionless structures, but an SI Preston term cannot become a public-data physics residual baseline merely by matching a number. Model comparison must remain native-unit and leakage-safe until a defensible unit/calibration bridge exists. Within that boundary, the tree supplies a strong offline point model, but the physics proxy and hybrid results are cautionary: structural plausibility does not guarantee predictive improvement when units and physical calibration are absent. The hybrid's significantly worse error blocks the intended physics-plus-residual claim rather than being hidden behind the tree result.

The public-data uncertainty and chronology findings also separate point accuracy from reliability. The primary tree's narrow symmetric intervals undercover both retained roles, and an extreme late-block target collapses chronological RMSE despite a small median error. Label and continuity sensitivities show that both coverage and tail behaviour depend on defensible data semantics. A controller or safety filter therefore cannot treat the WP09 interval as an approved operating bound, nor can it assume the random whole-wafer split establishes future temporal performance.

Finally, the study treats supervisory safety as an independent software contract, not as a property inferred from predictive accuracy. An accurate warning model can still propose an infeasible action, act on an invalid sensor, or resume too early. Keeping the safety filter separate enables explicit rejection, clipping, hold fallback, and restart tests. Those results remain future work.

13 Limitations and threats to validity

- The PHM archive has verified local provenance, but its embedded dataset licence is not separately stated. Its target unit is source-undeclared and process scaling factors are hidden.
- Process-mode proxies are inferred from inputs, 306 official wafer/stage groups remain unresolved across splits, four extreme training labels require preregistered sensitivity analyses, and all records use one machine ID.
- The source partitions reuse wafer IDs across roles; only the precedence-retained roles support the primary independent-wafer comparison. The selected tree's nominal 90% intervals cover only 83.9%/84.7%, and one known extreme label dominates the chronological stress RMSE.
- PHM data do not contain paired electrical/UPW utility histories and therefore cannot validate the proposed utility-to-CMP causal chain.
- UPS, VFD, motor, pump, UPW compliance, thermal, and sensor/network dynamics are reduced-order engineering approximations with synthetic capacities, coefficients, delays, and disturbances.
- The pump lacks manufacturer curves, efficiency maps, best-efficiency point, cavitation/NPSH, shaft loss at runout, and motor-load feedback. The hydraulic model omits distributed inertia and water hammer.
- The water-quality state is a dimensionless synthetic deviation, not measured chemistry or contamination. It has no CMP coupling.
- CMP uses spatially uniform pressure and reduced-order spindle, slurry, thermal, pad, wear, and dresser states. Numerical parameters are synthetic, not calibrated to a material/tool stack.
- The default temperature-to-MRR coefficient is zero. Thermal coupling is therefore a structural null for MRR in the default material profile.
- The reviewed literature supports pathway plausibility, not the target tool plumbing or numerical header thresholds. No-connection and zero-link cases remain scientifically necessary.
- Scenario labels and parameter distributions are synthetic definitions, not observed fab fault rates. Simulator truth supports internal scoring only.
- Primary WP12 TEST evidence has only three independent event-bearing runs. Positive endpoints require nearly full-DRESS synthetic service loss, and fixed severity grids are not samples from a fab distribution.
- Both learned warning models false-alarm severely under explicit no-connection and unseen-compound shifts. High synthetic noise damages discrimination, specificity, calibration, and empirical conformal coverage.
- Probability and conformal calibration use disjoint whole runs, but temporal dependence prevents a formal run-level guarantee. Empty prediction sets occur on 7.04% of logistic and 12.27% of gradient-boosted primary rows and require independent high-uncertainty handling.
- Known schedule, battery capacity, and load are simulator configuration; real availability is unverified. Reported WP12 timing is vectorized offline inference, not streaming controller latency.
- WP13 primary scoring supplies a neutral valid-positive warning, so its accuracy is conditional cause separability rather than an end-to-end warning-and-diagnosis rate.
- WP13 has only six TEST runs per class on a fixed synthetic severity grid. The grouped bootstrap resamples these runs but cannot create missing mechanisms or real-fab diversity.
- Rule only outperforms the frozen hybrid on TEST, so learned fusion has not established incremental value. The primary method is retained to avoid post-TEST selection, not because it is empirically superior.
- Pressure-sensor-fault recall is 0.3333. A 0.20 s communication delay forces all diagnoses to UNKNOWN, and 10% packet loss reduces known-cause coverage to 0.15. Diagnostic unavailability must not be treated as a confirmed normal state.
- Public-data virtual metrology is an offline source-native result only. Controller, safety-filter, integrated-runtime, and controller-robustness results are unavailable at this evidence freeze.
- No measured spatial metrology, physical-defect label, wafer-yield outcome, equipment trial, or real actuator interface is present. The work does not support defect, yield, equipment-damage, or production-control conclusions.
- A 10 ms synthetic timestep and eventual measured software latency must not be represented as a microsecond or production real-time guarantee.

14 Reproducibility and traceability

The Python environment is the project-local conda environment `devkki`. The WP08 reference trace contains 700 rows and has SHA-256 `902fa428...75626e`; its canonical runtime configuration hash is `99d7876e...e91670`. WP10 uses schema/interface versions 2.4.0/3.2.0 and runtime hash `b2b07edb...d7383f`. The frozen negative-control and positive-chain trace hashes are listed with their results above. The global sensitivity JSON records estimator method, seed, ranges, raw indices, and parameter provenance. WP12 uses experiment revision `1.4-no-test-selection`; its configuration hash is `b5373001...df1e05`, eligible-dataset hash is `fabe232a...b43fa`, and deterministic TEST/robustness probability payload hash is `5a4fb701...0ede`. The payload excludes measured latency and set-valued conformal outputs by design.

WP13 was opened once from clean Git commit `ec7bc8b41a...15a9540c` after 23 focused and 211 complete tests passed with warnings treated as errors. Its persistent opening-marker hash is `53139b8b...502a3d`; the 198-row, 66-run TEST payload hash is `395464f2...24329`. Validation JSON, deterministic payload, and prediction CSV hashes are `f075e513...13fefc`, `b1f7080d...e74f99`, and `8108f514...494eb`. The confusion and recall figure hashes are `07db0b4d...2b57ad` and `add6b0cf...11615`. The persistent marker prevents a silent second holdout execution.

WP09 was opened from clean Git checkpoint `01c59a6172...e4358805463c`. Its target-blind split payload hash is `44c88516...0ba944`; validation JSON and prediction CSV hashes are `52d25cf9...bd10c` and `ea3e9e81...79826`; and the deterministic validation payload hash is `84d4823b...320db`. All six model artifacts reload under checksum verification. The result records CPython, NumPy, pandas, Pydantic, PyYAML, and scikit-learn versions.

All six communication figures are generated from frozen CSV/JSON evidence. A machine-readable manifest records input and output SHA-256 values and states that it contains interim WP08/WP10 synthetic evidence only. This manuscript's two WP12 figures are generated by the frozen early-warning validation driver and have hashes `76863b51...53b16f` and `077177e0...2dcfcb`. Three WP09 figures are regenerated from the frozen JSON/CSV only; their input/output hashes and public-data claim boundary are recorded in a separate figure manifest. The two WP13 figures are generated by the one-shot attribution driver from the frozen held-out result. The evidence map links each quantitative statement to the governing report and artifact. The article is rebuilt locally with `conda run -n devkki make paper`; intermediate LaTeX files remain under `paper/build`, and the reviewed PDF is written under `reports/paper`.

Immediately before WP09 target access, 19 focused tests passed; the complete repository suite passed 192/192 with warnings treated as errors. These counts are historical evidence tied to the frozen revisions, not a claim that future changes automatically preserve the result. The analogous WP13 preflight counts are recorded above. Reproducible paired controller manifests and the final claims audit remain pending.

15 Conclusion

This work has established a bounded synthetic foundation for studying electrical and UPW disturbances in CMP. The utility plant conserves accepted UPS energy and UPW mass, the CMP subsystem respects process modes and stores consumable/thermal history, and a typed connection makes the assumed physical path explicit. The healthy-UPS sag remains a null, while a separately declared degraded-UPS dressing event yields a delayed 3.19% lower later simulated MRR through pad activity. Sensitivity results show that this magnitude depends materially on synthetic connection assumptions.

The detection half of the central question now has a narrow synthetic answer: two frozen models anticipate all three configured primary events with useful lead time. The structural-null and noise failures show why that result cannot be interpreted as topology-independent causality or permission to act. Conditional on a valid diagnostic warning, the frozen hybrid identifies 57/66 held-out initiating labels and abstains on every error; rule only identifies 61/66 and therefore remains an important stronger comparator. Communication delay and dropout sharply reduce diagnostic availability, and none of these simulator-label results constitutes experimental causal proof. Separately, a leakage-safe public-data tree predicts source-native average MRR well on retained whole-wafer holdouts, but interval undercoverage, a failed hybrid claim, and chronological tail fragility prevent stronger reliability or physical-model conclusions. The mitigation and safety half remains open until bounded control, independent safety filtering, and paired robust evaluation are complete. Supported conclusions remain limited to the retained PHM roles and to the tested synthetic equations, declared topology, and named WP12/WP13 ensembles.

Data, code, and manuscript status

The PHM archive was supplied locally and is not redistributed by the paper. The repository retains extraction checksums, preprocessing manifests, deterministic traces, figure hashes, tests, and the living evidence map. This is a journal-neutral draft; authorship, affiliations, funding, competing interests, acknowledgements, a target journal class, and submission-specific data/code availability language must be completed before submission.

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